

1 **Supporting Information**
2 **Assembly Processes and Ecological Network Patterns of Multitrophic Soil Mite**
3 **Communities in Karst Primary Forests**

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8 **The supporting information includes:**

9 **1)** Supplementary Text S1-S3

10 **2)** Supplementary Tables S1-S10

11 **3)** Supplementary Figures S1-S4

12

(1) Supplementary Text S1-S3

Text S1. Laboratory processing of soil biotic samples

Collection and Preservation of Soil Fauna. Soil fauna were extracted in the Soil Fauna Separation Laboratory of Guizhou Normal University using an improved Berlese-Tullgren funnel apparatus. Soil samples were placed in funnels above collection bottles containing 75% ethanol. The apparatus, housed in an incubation cabinet, was heated by a 40 W incandescent bulb at approximately 35°C for 48 hours until the soil dried completely. This gradient drove the fauna downward into the ethanol for preservation.

Sorting and Fixation of Soil Mites. The collected specimens were sorted under a stereomicroscope (Olympus SZX2-FOF, Japan). Mites were initially cleaned and fixed in 75% ethanol. Subsequently, they were treated differentially: transparent-bodied Gamasid mites were preserved long-term in 95% ethanol, whereas opaque Oribatid mites were cleared in 95% lactic acid at room temperature for approximately 15 days to facilitate identification.

Identification of oribatid and gamasid mite specimens. Mite specimens were prepared as temporary slides, and all adult individuals were morphologically identified and counted under an optical microscope (Olympus CX41RF, Japan). Identification was primarily based on the following references: *A Manual of Acarology* (Krantz and Walter, 2009), *Oribatid Mites: Biodiversity, Taxonomy and Ecology* (Behan-Pelletier & Lindo, 2023), *Pictorial Keys to Soil Animals of China* (Yin, 1998), *Soil Gamasid Mites in Northeast China* (Yin et al., 2013), *Acarology* (Li & Li, 1989), and *Biodiversity of Jilin Province. Fauna: Suborder Oribatida Volume* (Arachnida: Acari) (Liu et al., 2021). To ensure taxonomic accuracy and reliability, all specimens were identified only to the genus level. Larvae, nymphs, and damaged individuals were excluded from statistical analyses due to insufficient morphological characteristics. The use of genus-level identification as the taxonomic unit was adopted because it effectively reflects changes in soil mite diversity and community assembly mechanisms, an approach that has been widely validated (Meehan et al., 2019; Malloch et al., 2020; Zhou et al., 2022; Zhang et al., 2023; Chen et al., 2024). All identified voucher specimens are deposited in the Institute of Karst Studies, Guizhou Normal University.

Reference

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Text S2 Soil physicochemical characteristics measurements

Soil pH was measured potentiometrically in a 1:2.5 soil-to-water extract. Soil organic carbon (SOC) content was determined using the potassium dichromate oxidation method with external heating. Total nitrogen (TN) was measured by the Kjeldahl method following digestion. Total phosphorus (TP) and total potassium (TK) were analyzed after digestion by molybdenum-antimony anti-colorimetry and flame photometry, respectively. Based on the above results, soil carbon-to-nitrogen ratio (C/N) and carbon-to-phosphorus ratio (C/P) were calculated. All chemical analyses followed the standard methods described by Bao (2000). Soil bulk density (BD), natural water content (NWC), and saturated water content (SWC) were determined in situ using the cutting ring method. Soil total porosity (P) was calculated using the empirical formula $P = 93.947 - 32.995 \times BD$.

Reference

Bao, S.D. (2000) Soil and agricultural chemistry analysis. Beijing: Agriculture Publication. pp. 1-144 (In Chinese)

Text S3 Dtermining key soil environmental factors with multiple regression models

To quantify the relative effects of soil properties on the diversity and network complexity of mite multitrophic groups , this study employed a multi-model inference framework based on linear regression. This approach effectively addresses the common issue of predictor collinearity in ecological data and systematically evaluates the relative importance of each variable without relying on a single model.

First, a global linear model incorporating all predefined soil physicochemical factors was constructed. Predictors included: pH, porosity, bulk density, saturated water content, natural water content, soil organic matter, total nitrogen, total phosphorus, total potassium, and their stoichiometric ratios. To eliminate perfect multicollinearity, variables were screened for redundancy. Bulk density was found to have a perfect linear relationship with porosity and was therefore excluded from the global model. Subsequently, variance inflation factor (VIF) tests were conducted on the remaining predictors; all VIF values were below 10, indicating acceptable levels of multicollinearity. The normality of the model residuals was confirmed by the Shapiro-Wilk test.

Next, automated model selection was performed using the *dredge* function in package MuMIn, based on the small-sample corrected Akaike Information Criterion (AICc). This process systematically evaluated all possible subset models derived from the global model. Models were ranked according to their AICc values, and those with $\Delta AICc < 2$ were considered as the best-supported model set.

To fully account for model selection uncertainty, model averaging was applied to all models in the best-supported set. The *model.avg* function was used to calculate averaged parameter estimates. This study reports the full averages, which shrink the coefficients of non-significant predictors toward zero, thereby providing more robust estimates. Meanwhile, the relative importance of each predictor was assessed by summing the Akaike weights across all supported models in which it appeared.

(2) Supplementary Tables S1-S10

Table S1. R functions and packages used for statistical analyses.

Analysis	Function	Package	Reference	Note
1 Data cleaning	pivot_longer(), summarise()	tidyverse	1	Preparing data for visualization
2 Visualization	brewer.pal()	ggplot2 RColorBrewer	2 3	All statistical figure except network Select 'Set2' and 'Set1' color palettes
3 Regression Analysis	check_normality(), check_homogeneity(), check_heteroscedasticity()	performance	4	Check linear regression assumptions
	tab_model()	sjPlot	5	Statistical parameters of mixed-effects models
	vif()	car	6	Collinearity diagnostics
	ggpredict()	glmmTMB	7	Generate predictions with confidence intervals
	lmer()	lme4	8	Linear mixed effects model
	dredge(), subset(), model.avg()	MuMIn	9	Multiple linear model
	analyze_diversity()	performance, agricolae	4, 10	Between-group tests and multiple comparisons
4 Community Assembly	tNST(), dist.3col()	NST	11	Taxonomic normalized stochasticity ratio
	beta.div.comp()	adespatial	12	Beta diversity decomposition
	mite_nmds()	tidyverse, vegan	1, 13	NMDS and ANOSIM analysis
	community_mean_niche()	spaa	14	Levins' measure of niche breadth
5 Co-occurrence network	extract_sample_net()	MetaNet	15	Obtaining topological parameters of subsample
	net()	Hmisc, igraph	16, 17	Network based on significant Spearman correlations
	process_net()	igraph	17	Coexisting network topological attributes
6 Alpha diversity	alpha_diversity1()	vegan	13	Genus richness and Shannon index
7 Min-Max normalization	normalize()	tidyverse	1	Composite indicator by averaging
8 Pearson correlation analysis	correlation()	pheatmap	18	Clustering heatmap
9 Structural equation modeling	as.psem(), update()	piecewiseSEM	19	Direct and indirect effects of network complexity

Notes: This study utilized the citation() function in R to retrieve reference information and employed the packageVersion() function to determine the software package version. Double underscores denote user-defined functions.

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157 **Table S2. Alpha diversity of mite groups across multiple trophic levels in the karst forest.**

Forest sites	Total mites			Fungal feeders			Predators			Primary decomposers		
	abd	rich	H'	abd	rich	H'	abd	rich	H'	abd	rich	H'
RBS_1	450	35	2.55	294	9	1.38	111	18	2.47	43	6	1.73
RBS_2	613	47	3.00	398	19	2.02	102	16	2.47	86	10	1.89
RBS_3	493	28	2.05	396	11	1.31	66	11	2.05	18	4	1.21
RBS_4	1324	38	2.55	1039	16	2.00	126	14	2.16	149	6	0.60
RBS_5	940	31	2.59	748	14	2.14	78	14	2.25	114	3	0.46
RBS_6	1552	47	2.81	1187	17	2.14	171	21	2.58	175	7	1.11
DBF_7	505	40	2.79	335	12	1.85	119	20	2.46	43	6	1.02
DBF_8	190	33	2.91	120	12	1.98	45	16	2.53	18	3	0.65
DBF_9	539	45	3.11	354	14	2.11	168	26	2.99	16	4	1.21
DBF_10	1286	61	3.11	824	23	2.17	364	31	2.89	91	6	0.64
DBF_11	908	52	3.14	604	19	2.29	215	27	2.93	83	5	0.65
DBF_12	734	49	2.90	472	21	1.98	164	23	2.76	96	4	0.88
EBF_13	913	51	3.00	524	18	1.95	229	23	2.68	140	8	1.06
EBF_14	116	19	2.49	51	8	1.71	55	9	1.60	10	2	0.61
EBF_15	684	53	3.08	423	18	2.09	142	24	2.79	112	9	1.46
EBF_16	618	71	3.74	332	34	2.92	240	26	2.94	45	10	1.80
EBF_17	384	54	3.41	206	25	2.59	147	22	2.68	30	6	1.05
EBF_18	422	61	3.50	256	29	2.83	121	30	3.00	45	2	0.18
CBF_19	1153	63	3.34	699	23	2.27	293	31	3.02	140	7	1.66
CBF_20	792	59	3.27	441	20	2.06	233	29	3.04	90	8	1.50
CBF_21	1322	68	3.44	713	31	2.44	446	28	2.81	136	7	1.44
CBF_22	847	58	3.38	532	24	2.62	216	23	2.50	88	9	1.56
CBF_23	524	54	3.43	302	23	2.60	151	24	2.73	46	6	1.25
CBF_24	916	63	3.29	612	26	2.54	188	27	2.59	107	9	1.46

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159 **Notes:** Abundance (abd), richness (rich), and Shannon diversity (H').

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Table S3. Total mite abundance and species richness across various taxa and forest types.

Category	Abundance (ind.)	Richness (genus)
Total mites	18225	135
Oribatida	14779	91
Mesostigmata	3446	44
Fungal feeders	11862	54
Predators	4190	57
Primary decomposers	1921	21
<u>Taxa with mineralized cuticle</u>	56	4
Lichen feeders	158	2
Moss feeders	94	1
RBS	5372	74
DBF	4162	84
EBF	3137	105
CBF	5554	105

Notes: Taxa with mineralized cuticles (indicated by underlining) were categorized as fungal feeders/secondary decomposers due to their incorporation of CaCO₃ for defense. Moss feeders and lichen feeders (indicated by strikethrough) were excluded from this study due to their low species representation. Riparian shrubland (RBS), deciduous broadleaf forest (DBF), evergreen broadleaf forest (EBF), and mixed coniferous-broadleaf forest (CBF).

Table S4. ANOSIM analysis of the multitrophic community composition of mites in different forest types.

Group1	Group2	R	Significance	R	Significance
Paired groups		Total mites		Fungal feeders	
RBS	DBF	0.378	0.004 (**)	0.352	0.004 (**)
RBS	EBF	0.437	0.004 (**)	0.406	0.008 (**)
RBS	CBF	0.596	0.005 (**)	0.443	0.005 (**)
DBF	EBF	0.115	0.157 (ns)	0.168	0.064 (ns)
DBF	CBF	0.339	0.018 (*)	0.357	0.025 (*)
EBF	CBF	0.191	0.068 (ns)	0.154	0.083 (ns)
		Predators		Primary decomposers	
RBS	DBF	0.154	0.124 (ns)	0.608	0.007 (**)
RBS	EBF	0.180	0.087 (ns)	0.256	0.055 (ns)
RBS	CBF	0.507	0.005 (**)	0.380	0.018 (*)
DBF	EBF	0.082	0.184 (ns)	0.215	0.083 (ns)
DBF	CBF	0.250	0.015 (*)	0.163	0.083 (ns)
EBF	CBF	0.326	0.01 (*)	0.317	0.028 (*)

Notes: Analysis of similarities (ANOSIM) was performed on a Bray-Curtis distance matrix (n = 6) to test for differences in mite community composition among forest types. The R-statistic (range: -1 to 1) reflects the degree of separation between groups; values near 1 indicate strong dissimilarity. Significance was assessed with 999 permutations, and levels are denoted as: ***p < 0.001, **p < 0.01, *p < 0.05 (bolded); ns, not significant.

Table S5. Relationships among different mite trophic groups across alpha diversity, beta diversity, and network complexity levels.

Alpha diversity	Dependent variables	Fungal feeders					
		Intercept	Slope	95% CI	Marginal R ²	Conditional R ²	p-value
	Predators	0.33	0.66	[(0.15 - 0.77), (0.56 - 1.22)]	0.397	0.528	0.003
	lmer (alpha_diversity_predator~ alpha_diversity_fungi + (1 Site), data)						
	Dependent variables	Primary decomposers					
		Intercept	Slope	95% CI	Marginal R ²	Conditional R ²	p-value
	Predators	0.50	0.27	[(0.23 - 0.52), (0.78 - 1.04)]	0.075	0.397	0.164
	lmer (alpha_diversity_predator~ alpha_diversity_decomp + (1 Site), data)						
Beta diversity	Dependent variables	Fungal feeders					
		Intercept	Slope	95% CI	Marginal R ²	Conditional R ²	p-value
	Predators	0.00	0.69	[(-0.85 - 0.42), (-0.40 - 0.88)]	0.708	0.801	0.000
	lmer (beta_predator ~ beta_fungi + (1 Site), data)						
	Dependent variables	Primary decomposers					
		Intercept	Slope	95% CI	Marginal R ²	Conditional R ²	p-value
	Predators	0.00	-0.12	[(-0.49 - -0.23), (0.23 - 0.47)]	0.027	0.315	0.464
	lmer (beta_predator ~ beta_decomp + (1 Site), data)						
Network complexity	Dependent variables	Fungal feeders					
		Intercept	Slope	95% CI	Marginal R ²	Conditional R ²	p-value
	Predators	0.23	0.71	[(0.26 - 0.62), (0.49 - 0.94)]	0.452	0.706	0.000
	lmer (complexity_predator ~ complexity_fungi + (1 Site), data)						
	Dependent variables	Primary decomposers					
		Intercept	Slope	95% CI	Marginal R ²	Conditional R ²	p-value
	Predators	0.44	0.14	[(0.35 - 0.41), (0.62 - 0.72)]	0.01	0.298	0.598
	lmer (complexity_predator ~ complexity_decomp + (1 Site), data)						
	Dependent variables	Primary decomposers					
		Intercept	Slope	95% CI	Marginal R ²	Conditional R ²	p-value
	Fungal feeders	0.29	0.18	[(0.20 - 0.26), (0.50 - 0.61)]	0.013	0.426	0.491
	lmer (complexity_fungi ~ complexity_decomp + (1 Site), data)						

Notes: Displayed are the intercept, slope, and 95% confidence interval of the fitted line from the linear mixed-effects model, with associated conditional R², marginal R², and p-value.

Table S6. Partitioning of total beta diversity (BDtotal) into its turnover (Repl) and nestedness (RichDif) components for soil mite multitrophic groups.

Feeding Guild	BDtotal	Repl	RichDif	Repl/Bdtotal	RichDif/Bdtotal
Total mites	0.286	0.169	0.117	59.14%	40.86%
Fungal feeders	0.266	0.118	0.148	44.33%	55.67%
Predators	0.278	0.155	0.123	55.78%	44.22%
Primary decomposers	0.362	0.211	0.152	58.10%	41.90%

Notes: Total Beta Diversity (BDtotal) quantifies the overall compositional dissimilarity of species between forest sites. The Turnover component (Repl) represents the proportion of beta diversity driven by species replacement. In contrast, the Nestedness component (RichDif) represents the proportion attributable to differences in species richness.

Table S7. Classification of potential species interaction types within and between trophic groups.

Association types	Potential species interactions	Edge number					Abbreviation
		Total mites	RBS	DBF	EBF	CBF	
Positive within-trophic	Fungal feeders ↔ Fungal feeders (+)	122	31	52	71	44	F-F (+)
	Predators ↔ Predators (+)	80	35	79	65	45	P-P (+)
	Primary decomposers ↔ Primary decomposers (+)	11	7	4	15	2	D-D (+)
	Fungal feeders ↔ Primary decomposers (+)	61	22	36	56	21	F-D (+)
Negative within-trophic	Fungal feeders ↔ Fungal feeders (-)	8	9	2	7	25	F-F (-)
	Predators ↔ Predators (-)	2	9	4	24	26	P-P (-)
	Primary decomposers ↔ Primary decomposers (-)	1	0	4	1	1	D-D (-)
	Fungal feeders ↔ Primary decomposers (-)	0	9	12	5	21	F-D (-)
Positive cross-trophic	Predators ↔ Fungal feeders (+)	191	55	130	129	75	P-F (+)
	Predators ↔ Primary decomposers (+)	40	33	44	55	31	P-D (+)
Negative cross-trophic	Predators ↔ Fungal feeders (-)	15	29	10	26	48	P-F (-)
	Predators ↔ Primary decomposers (-)	1	7	9	8	14	P-D (-)

Table S8. Topological parameters of ecological networks in the mite meta-community and local communities of different forest types.

Topological parameters	Meta-community	Feeding Guild			Forest type			
	Total mites	Fungal feeders	Predators	Primary decomposers	RBS	DBF	EBF	CBF
Total nodes	132	54	57	21	72	82	102	103
Total links	532	130	82	12	246	385	463	359
Positive links	505	122	80	11	183	344	392	224
Negative links	27	8	2	1	63	41	71	135
Average degree	8.06	4.81	2.88	1.14	6.83	9.39	9.08	6.97
Network density	0.0615	0.0908	0.0514	0.0571	0.0962	0.1159	0.0899	0.0683
Network diameter	4.50	4.59	3.62	4.15	5.05	8.49	9.20	8.55
Average path length	1.73	1.90	1.58	1.93	1.98	2.58	3.03	3.09
Average clustering coefficient	0.4012	0.3945	0.4030	0.0000	0.5851	0.7451	0.6233	0.6059
Betweenness centrality	0.0570	0.1524	0.0702	0.1274	0.0445	0.0854	0.1361	0.1034
Modularity	0.4838	0.5144	0.6364	0.5605	0.4828	0.4597	0.5865	0.5970

Table S9. Best-fitting models predicting alpha/beta diversity and network complexity of multi-trophic mite groups.

	Predictors	Estimate	95% CI	p-value	R ²	Adjusted R ²	N	AICc range
Alpha diversity of fungal feeders	C:N	0.118	[0.05 - 0.19]	0.002 **	0.698	0.634	1	-11.7
	TN	-0.084	[-0.15 - -0.01]	0.021 *				
	TK	-0.128	[-0.20 - -0.05]	0.002 **				
	SWC	-0.160	[-0.23 - -0.09]	0.000 ***				
Alpha diversity of predators	C:N	0.006	[-0.11 - 0.16]	0.808	0.388	0.219	7	[3.0 - 4.7]
	pH	-0.027	[-0.18 - 0.08]	0.567				
	TK	-0.140	[-0.32 - 0.18]	0.011 *				
	Por	-0.027	[-0.32 - 0.18]	0.606				
Alpha diversity of primary decomposers	SWC	-0.050	[-0.32 - 0.18]	0.421	0.277	0.169	3	[6.9 - 8.5]
	C:N	0.102	[-0.01 - 0.27]	0.175				
	TK	0.047	[-0.04 - 0.22]	0.462				
	pH	-0.020	[-0.15 - 0.11]	0.667				
Beta diversity of fungal feeders	C:N	0.204	[0.06 - 0.35]	0.007 **	0.668	0.598	1	21.2
	TN	-0.194	[-0.43 - -0.15]	0.008 **				
	TK	0.256	[-0.41 - -0.11]	0.001 **				
	SWC	-0.287	[-0.43 - -0.15]	0.000 ***				
Beta diversity of predators	C:N	0.021	[-0.05 - 0.23]	0.677	0.498	0.359	5	[16.1 - 17.9]
	TN	-0.014	[-0.23 - 0.05]	0.740				
	TK	-0.216	[-0.37 - -0.06]	0.003 **				
	Por	-0.068	[-0.37 - 0.30]	0.486				
Beta diversity of primary decomposers	SWC	-0.108	[-0.48 - 0.16]	0.281	0.723	0.625	7	[23.5 - 25.1]
	TN	0.271	[0.09 - 0.46]	0.002 **				
	pH	0.042	[-0.21 - 0.45]	0.591				
	TP	0.027	[-0.36 - 0.33]	0.676				
Network complexity of fungal feeders	SWC	0.221	[-0.25 - 0.64]	0.133	0.704	0.599	3	[-47.5 - -46.1]
	NWC	0.271	[-0.12 - 0.42]	0.423				
	Por	0.020	[-0.36 - 0.40]	0.753				
	C:N	0.067	[0.05 - 0.17]	0.000 ***				
Network complexity of predators	TP	0.006	[-0.15 - -0.03]	0.673	0.447	0.294	4	[-23.9 - -22.5]
	TK	-0.046	[-0.15 - -0.03]	0.014 *				
	TN	-0.059	[-0.15 - -0.03]	0.001 **				
	NWC	0.011	[-0.05 - 0.12]	0.599				
Network complexity of primary decomposers	SWC	-0.072	[-0.19 - -0.04]	0.001 **	0.167	0.088	3	[-36.6 - 35.4]
	C:N	0.012	[-0.04 - 0.11]	0.623				
	pH	-0.008	[-0.09 - 0.06]	0.693				
	TK	-0.080	[-0.15 - -0.01]	0.009 **				
Network complexity of primary decomposers	NWC	-0.010	[-0.09 - 0.07]	0.697	0.167	0.088	3	[-36.6 - 35.4]
	SWC	-0.057	[-0.14 - 0.02]	0.124				
	C:N	0.021	[-0.00 - 0.09]	0.405				
	TK	0.007	[-0.02 - 0.08]	0.677				

Notes: Model averaging was used to identify the best subset of predictors; their parameter estimates and p-values are reported. The R², adjusted R², and 95% confidence intervals are from the final model incorporating these predictors. Model uncertainty is indicated by N (the number of models in the best subset) and their AICc range.

Table S10. Summary of the piecewise structural equation model (SEM) for network complexity in multi-trophic mite groups.

Feeding Guild	Response	Predictor	Estimate	Std. Error	DF	Std. Estimate
Fungal feeders	Soil properties	Geospatial factors	0.212	0.082	22	0.483 *
	α -diversity	Geospatial factors	0.130	0.111	21	0.162
	α -diversity	Soil properties	1.349	0.253	21	0.739 ***
	β -diversity	Geospatial factors	0.244	0.218	21	0.161
	β -diversity	Soil properties	2.508	0.497	21	0.726 ***
	Network complexity	Soil properties	0.459	0.200	21	0.385 *
	Network complexity	α -diversity	0.362	0.109	21	0.554 **
	$\sim\alpha$ -diversity	$\sim\beta$ -diversity	0.879		24	0.878 ***
Predators	Soil properties	Geospatial factors	0.238	0.049	22	0.720 ***
	α -diversity	Geospatial factors	0.275	0.202	21	0.327
	α -diversity	Soil properties	0.928	0.613	21	0.364
	β -diversity	Geospatial factors	-0.223	0.177	20	-0.191
	β -diversity	Soil properties	1.053	0.540	20	0.298
	β -diversity	α -diversity	1.118	0.183	20	0.804 ***
	Network complexity	Soil properties	0.534	0.275	21	0.357
	Network complexity	β -diversity	0.534	0.078	21	0.483 *
Primary decomposers	α -diversity	Soil properties	3.310	1.149	22	0.523 **
	β -diversity	Soil properties	-2.275	2.714	22	-0.175
	Network complexity	α -diversity	0.287	0.055	22	0.741 ***
	$\sim\alpha$ -diversity	$\sim\beta$ -diversity	0.151		24	0.15

(3) Supplementary Figures S1-S4

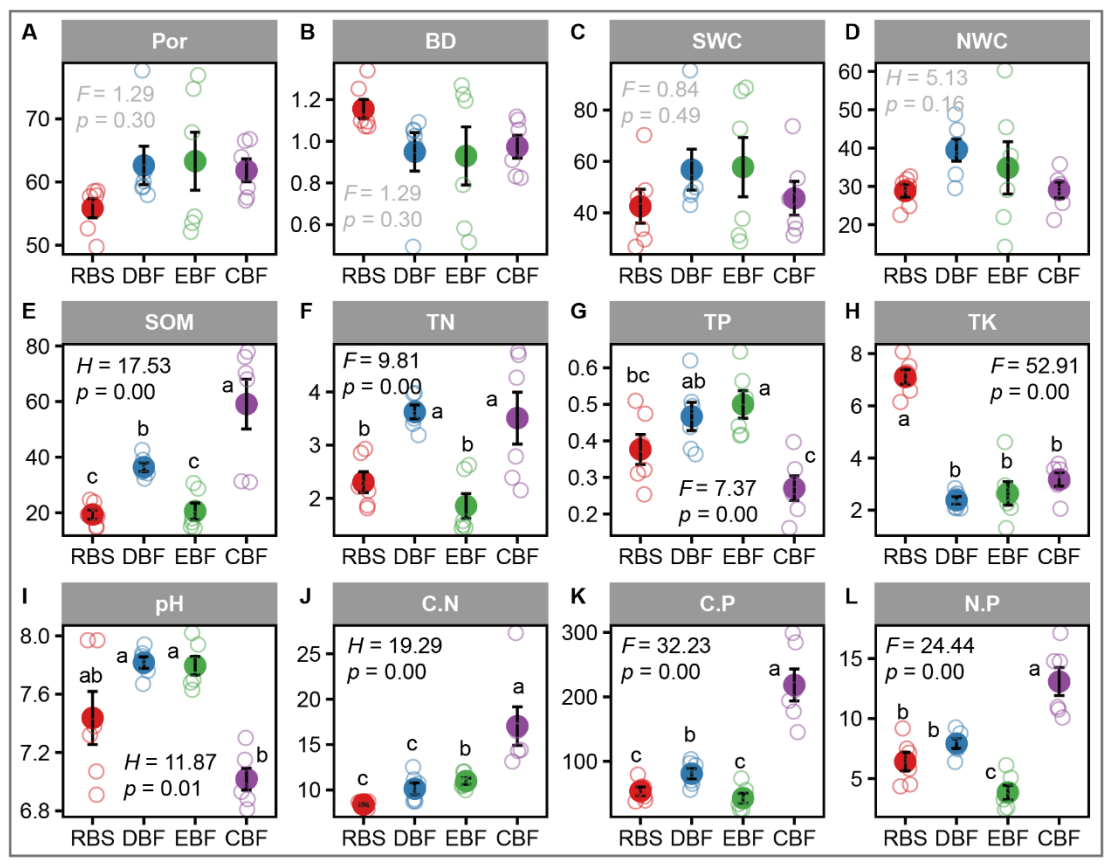


Figure S1. Soil properties of karst forests.

Notes: Dotted lines represent the mean $\pm$ standard error (SE). Lowercase letters indicate significant differences ($p < 0.05$) among sites. Data that met assumptions of normality and homoscedasticity were analyzed by linear regression followed by LSD post-hoc tests; otherwise, the Kruskal-Wallis test with Mann-Whitney U post-hoc tests was applied. P-values were adjusted using the Bonferroni method. Abbreviations are as follows: soil organic carbon (SOC); total nitrogen (TN); total phosphorus (TP); total potassium (TK); natural water content (NWC); saturated water content (SWC); bulk density (BD); porosity (Por); C:N, C:P, and N:P stoichiometric ratios; and pH.

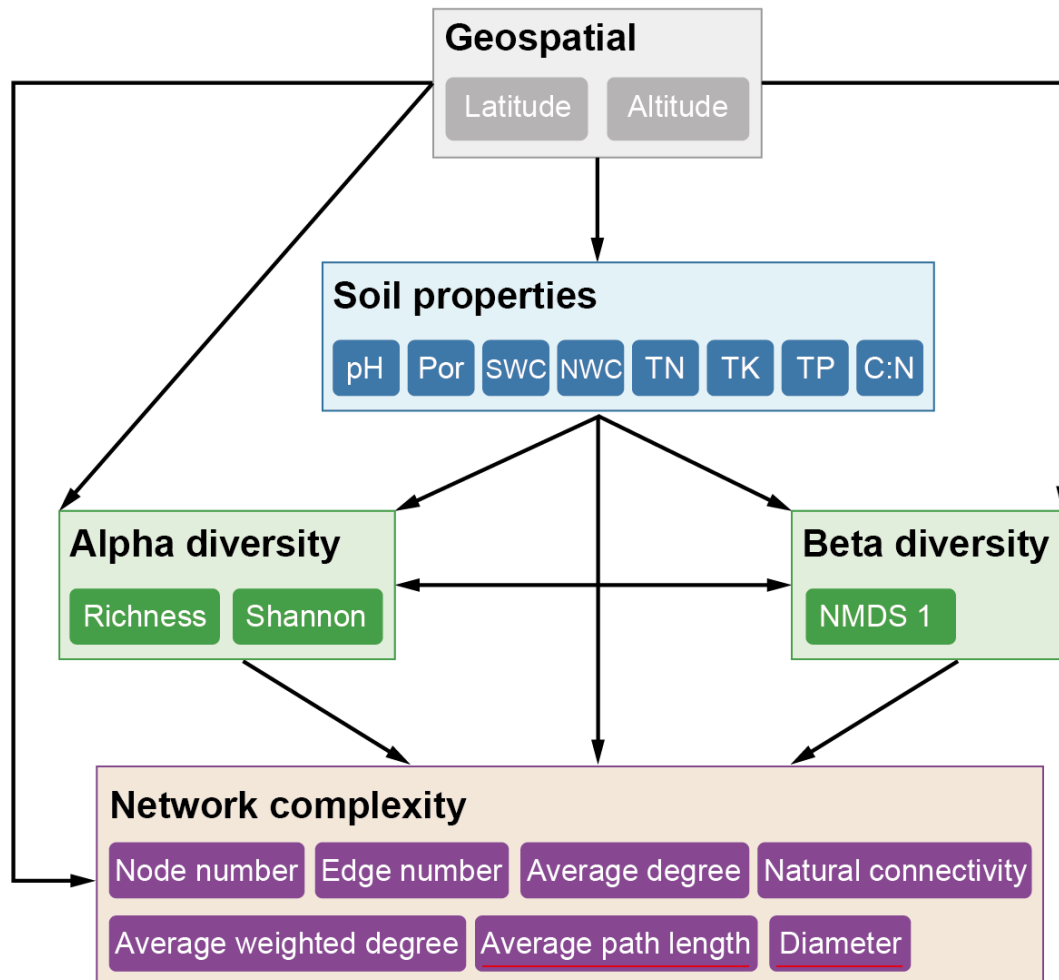


Figure S2. A priori model in structural equation modeling (SEM).

Notes: This a priori structural equation model (SEM) depicts the hypothesized relationships among geospatial factors, soil properties, diversity, and their potential effects on mite network complexity. Abbreviations for soil explanatory variables are defined in Figure S1.

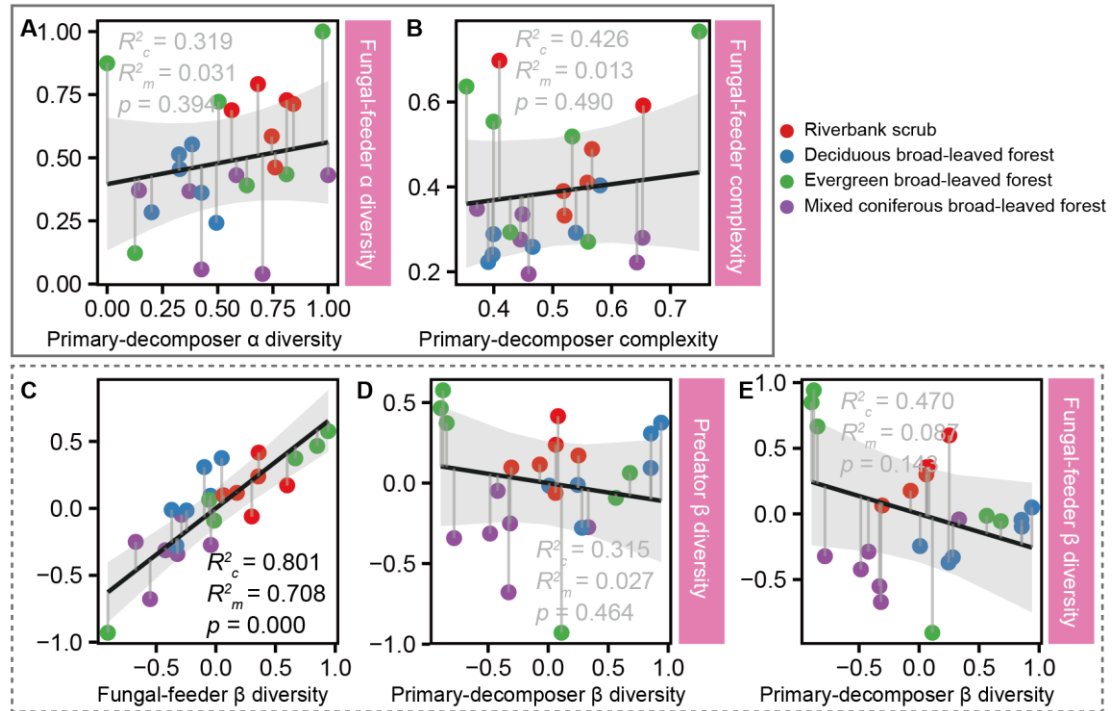


Figure S3. Relationships among mite trophic groups across metrics of alpha diversity (A), network complexity (B), and beta diversity (C-E).

Notes: Points represent individual soil samples ($n = 24$), colored by forest type ($n = 6$ samples per type). The solid black line shows the fit of a linear mixed-effects model, with the shaded area denoting the 95% confidence interval. The conditional R^2 (R^2_c) represents the variance explained by both fixed and random effects, while the marginal R^2 (R^2_m) indicates the variance explained by fixed effects alone.

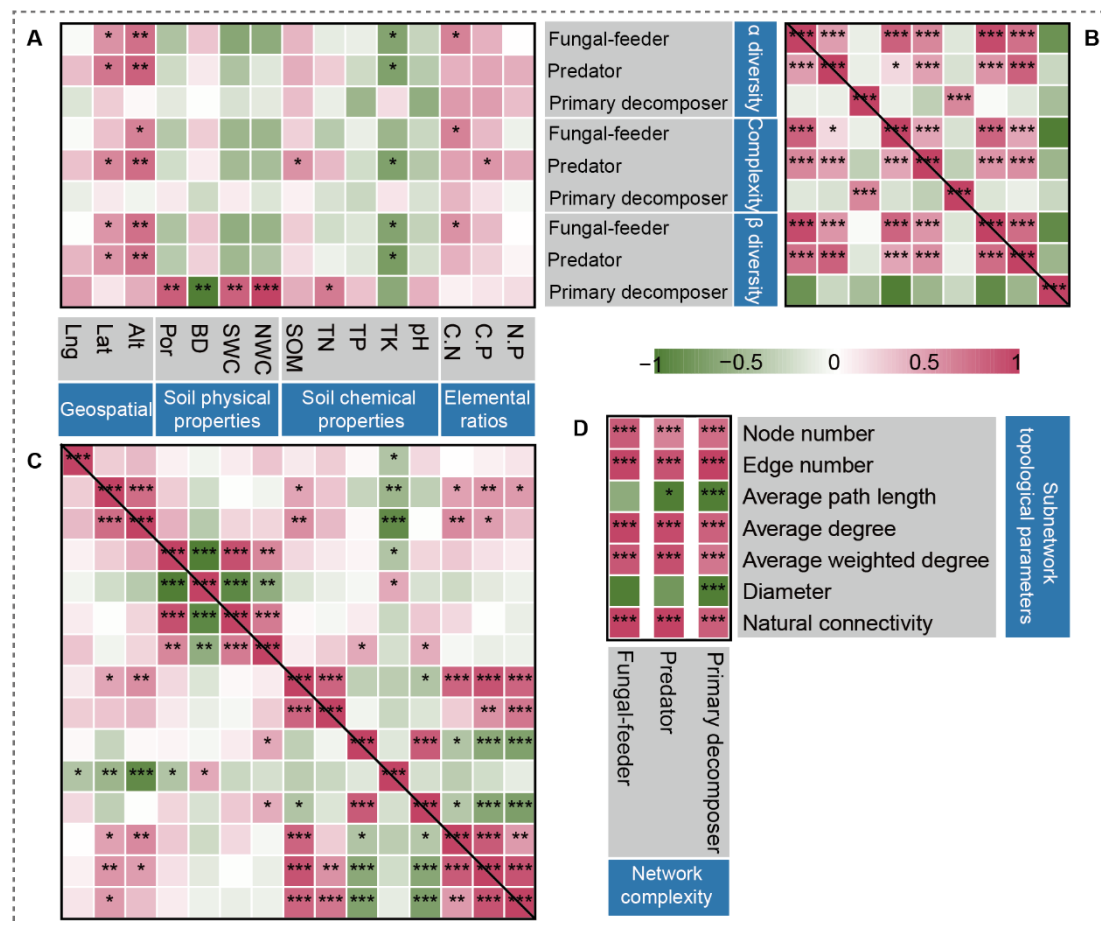


Figure S4. Pearson correlations between mite community properties and environmental factors, and among variables within these groups.

Notes: A, Matrix of Pearson correlations between alpha and beta diversity, network complexity of mite trophic groups, and environmental factors. B, Matrix of Pearson correlations among alpha diversity, beta diversity, and network complexity of mite trophic groups. C, Matrix of Pearson correlations among geospatial factors, soil physical properties, soil chemical properties, and ecological stoichiometry. D, Matrix of Pearson correlations between network complexity and individual variables. Positive and negative correlations are shown in red and green, respectively. Significance levels: *P < 0.05, **P < 0.01, ***P < 0.001. Abbreviations for soil properties are defined in Figure S1.